



RAPID ASSESSMENT AND RECOMMENDATIONS FOR ECOTOURSIM DEVELOPMENT IN THE HACIENDA SANTA MÓNICA

Coclé, Republic of Panama



Pristine beach at the Juan Hombrón Unit.

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RAPID ASSESSMENT AND RECOMMENDATIONS FOR ECOTOURSIM DEVELOPMENT IN THE HACIENDA SANTA MÓNICA

Coclé, Republic of Panama

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In cooperation with:

ANCON

Hacienda Santa Mónica

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Dead snags provide perching habitat for birds and great photography opportunities.

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I. INTRODUCTION

Grupo Verde Azul (GVA), managers of Hacienda Santa Monica (HSM) in Coclé Province, requested assistance from ANCON and the US Forest Service for a rapid assessment of the recreational potential at this property. The HSM property is a fully functional farm producing mostly beef cattle, but there is also interest in expansion to include produce (vegetables), dairy, and related agricultural products. More recently, GVA became interested in developing nature-based recreational opportunities on portions of this property (Map 1).

This report is a summary of a very preliminary, rapid assessment with recommendations for recreational opportunities on this property. Much more field reconnaissance is needed to develop a more in-depth assessment and plans.

II. FIELDWORK

A US Forest Service team visited the HSM and surrounding properties on 20-21 March 2015 to conduct a rapid assessment and to gather data on opportunities for ecotourism development in the area (Map 1). Under the direction of Jerry Bauer, the assessment team was comprised of Jerry Bauer, Jerry Wylie, Wayne Arendt, and Benny Wilson. Graphic designs were developed by Jorge Paniagua.

A briefing/introduction meeting was held with GVA staff and ANCON on 17 March in its Panama City office, with a subsequent short site visit made to the HSM.

III. ENVIRONMENTAL SETTING

The HSM is a large, single-owner tract of land located just west of the Río Hato community on Panama's Pacific Coast (Map 1). The HSM is comprised of two large tracts of land which total about 2,800 hectares.

The Hacienda Unit is located in a tropical savanna¹ ecosystem extending from the Pan-American Highway to a small section of the Pacific Coast. The Hacienda Unit has been under continual agricultural use for more than a century, mostly for beef cattle production. Ecologically it

¹ A savanna is "...a natural and stable ecosystem occurring under a tropical climate, having a relatively continuous layer of xeromorphic grasses and sedges, and often with a discontinuous layer of low trees and shrubs." (Kirchner, 1977, cited in Beard 1953).



Scouting.



Undeveloped trails provide access for visitors.



Beach at Juan Hombrón.

is located in the largest Neotropical savanna ecosystem in Panama (<http://ntsavanna.com>). Now a highly disturbed site, it is believed that this area still has remnant patches of the original tropical savanna.

The Juan Hombrón Unit, a much smaller tract of land, is located along the Pacific coast and is dominated by a mangrove ecosystem². This unit is less disturbed and is comprised of mangrove forest (mostly black, but also areas of red mangroves), some of which are in very good condition with older, large trees. Due to time limitations, the assessment team was only able to physically visit a very small section of the Juan Hombrón Unit.

IV. GOALS AND OBJECTIVES

The general goals and objectives of sustainable tourism are to maximize the positive benefits that tourism can have on biodiversity, ecosystems, and economic and social development, while minimizing negative social and environmental impacts from tourism (Secretariat of the Convention on Biological Biodiversity, 2004). In addition,

² A mangrove ecosystem is comprised of an assemblage of tropical trees and shrubs that inhabit the coastal intertidal zone. A mangrove community is composed of plant species whose special adaptations allow them to survive the variable flooding and salinity stress conditions imposed by the coastal environment." (Syste 2014).

an important goal is to maintain a high level of biodiversity as an attraction for tourism.

GVA could establish the following goals and objectives for this site:

Goals

- Conserve biodiversity, especially birds, other wildlife, and mangroves
- Change public attitudes and behaviors relating to mangroves
- Provide low-impact outdoor recreation to local and international tourists
- Create sustainable benefits for the environment, private sector, and visitors
- Diversify economic benefits from the hacienda by offering nature-based tourism

Objectives

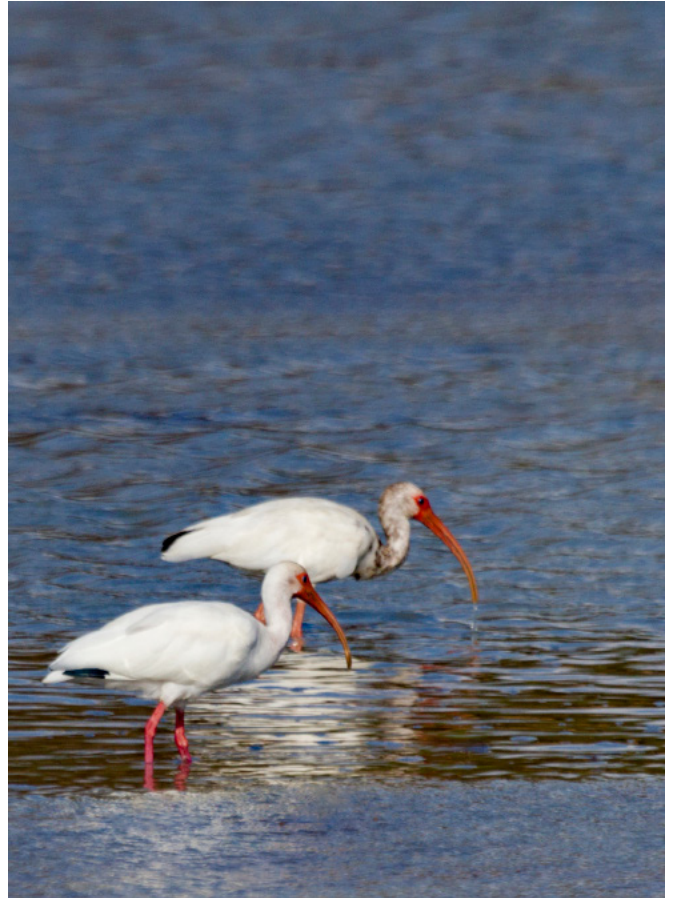
- Provide high-quality environmental education and interpretation
- Provide incentives for the private sector
- Develop proper infrastructure and signage
- Develop bird conservation programs, especially migratory shorebirds
- Create an identity for the Hacienda Santa Monica as a destination



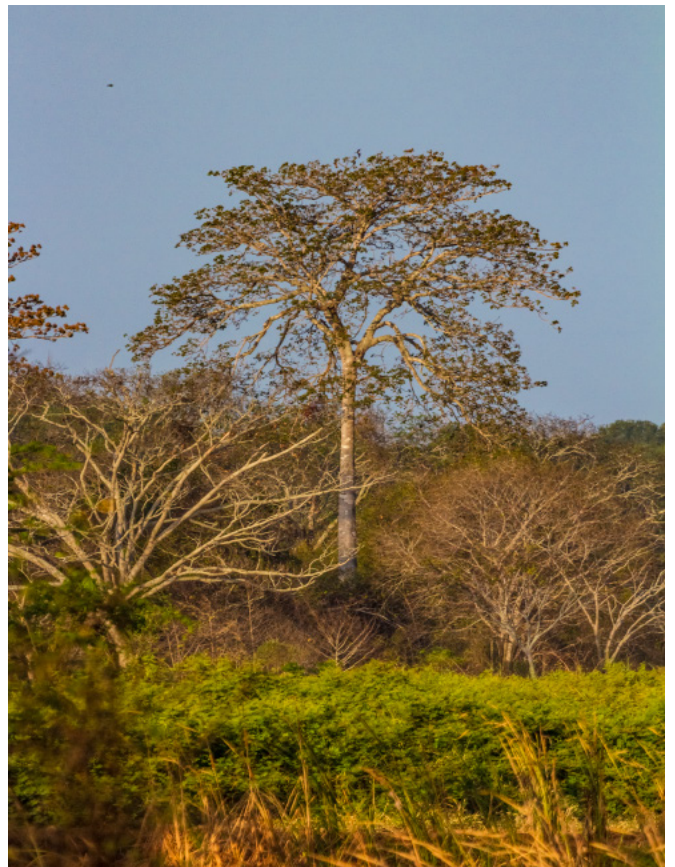




Jerry Wylie scouting coast-forest edge.



White ibis feeding in lagoon.



Dry forest near stream.

V. A BENEFITS BASED APPROACH

Conservation must be integrated with development activities. However, the ultimate goal of integrating sustainable tourism and biodiversity conservation is not just to reduce threats or avoid impacts, it is the creation of sustainable benefits to the environment and the local community (Global Sustainable Tourism Criteria, 2009; USAID/ARD 2005:28) and visitors, which is frequently overlooked.

To achieve this integration and meet the project objectives listed above requires a benefits-based model of sustainability that provides linkages and synergies between all three elements. Since they are directly linked, actions in one affect the other two (Figure 1). Details of this model are presented in Appendix A (Wylie and Bauer, 2008).

This model is not only a theoretical concept, it is a practical “compass” for providing direction at each stage of project planning, implementation and monitoring. For example, it can help:

- Define and expand the project vision and goals
- Focus the threats and opportunity assessments
- Develop integrated conservation/tourism targets
- Analyze positive “value-chain” linkages affecting tourism opportunities
- Create a conceptual model describing linkages among causal factors
- Develop specific actions with monitoring measures
- Select and prioritize activities with synergistic linkages

Figure 1.
Sustainable Tourism Model.



VI. SWOT ANALYSIS

Table 1 shows the SWOT analysis conducted during this rapid assessment. This is based on one field visit accessing the area via local, low-volume dirt roads and by hiking parts of the Hacienda. The SWOT is divided into two parts, 1) the Hacienda unit which is mostly a tropical savanna and the primary cattle raising/farming area, and 2) the Juan Hombrón Unit, which includes well-developed mangrove forest, estuary and large stretches of pristine pacific coast beaches.

The area’s primary attractions are large size, flagship bird and wildlife species, its pristine mangrove forests (mostly red and

black mangrove), as well as wildlife viewing opportunities for birds and other wildlife. These natural attractions are enhanced by their close proximity to the Pacific Coast where major tourism developments are underway.

Thousands of potential tourists (both local and international) are within a few kilometers of HSM. This potential source of visitors, in such close proximity, provides for a tremendous opportunity to develop ecotourism products to supplement the typical “sun-sand-and-sea” resort experience that is being marketed at the nearby luxury hotels. These opportunities are described in detail below.

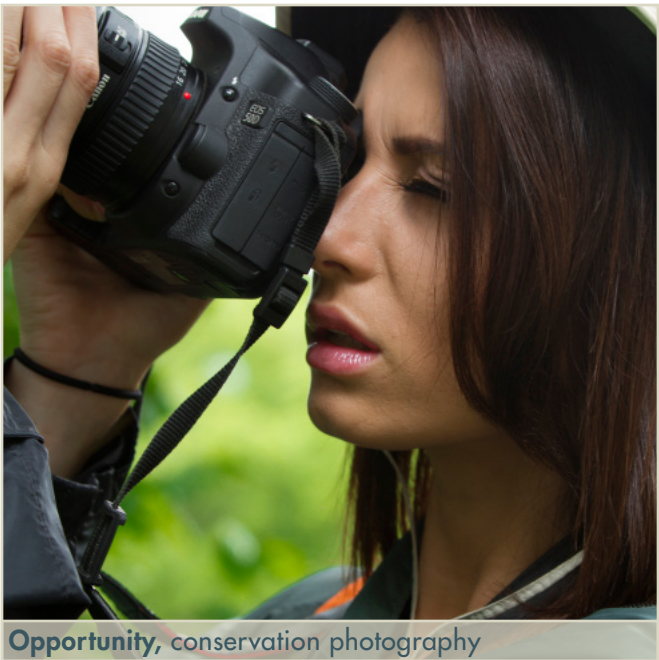


Table 1. Strengths, Weaknesses, Opportunities, Threats (SWOT) Rapid Assessment.

HACIENDA UNIT	
Strengths	
• Flagship bird species • Other wildlife, including large mammals • Last remnant savanna in Panama • Large size (2,800 hectares) • Excellent access • Relatively close to Panama (2 hours) • Major international hotels nearby (15 minutes) • Close to El Valle de Antón (about 45 minutes) • International airport nearby	
Weaknesses	
• Lack of trees and shade • Industrial-scale cattle raising • Unattractive infrastructure • Limited infrastructure • Uncomfortably hot in dry season • Caiman in ponds	
Opportunities	
• Conservation photography ranch tourism (new niche market emphasizing photography on private lands) • Soft adventure tourism (mountain biking, 4-wheeling, horseback riding) • Other potential private properties adjacent	
Threats	
• Pesticides from rice farming • Uncontrolled hunting • Environmental damage from cattle grazing	

JUAN HOMBRÓN UNIT	
Strengths	
• Flagship bird species • Other wildlife, including large mammals • Large size (? hectares) • Habitat diversity including lagoon/estuary • Well-developed mangroves • Attractive dry forest (guanacaste and cotui) • Native grasses • Extensive pristine beach • Very good access • Relatively close to Panama (2 hours) • Major international hotels nearby (15 minutes) • Close to El Valle de Antón (about 45 minutes) • International airport nearby	
Weaknesses	
• Uncomfortable in dry season • Lagoon is dry and unattractive in dry season	
Opportunities	
• Conservation photography ranch tourism (new niche market emphasizing photography on private lands) • Soft adventure tourism (mountain biking, 4-wheeling, horseback riding) • Environmental education • Kayak tours in mangrove channels • Soft adventure tourism • Beach activities • Saltwater fishing • Luxury tent ecolodge • Native savanna grass	
Threats	
• Water contamination • Invasive grasses • Caiman • Sea level rising	



Forested area where river and sea meet.

VII. PRELIMINARY CONCLUSIONS

While final conclusions must await the results of further fieldwork, preliminary observations suggest that tourism activities in the study area may be developed for specific niche markets, such as:

Recreational

- Bird-watching with a trained guide (Appendix B)*
- Walking and sightseeing/photography
- Conservation ranch photography (Appendix C)*
- Mountain biking
- Horseback riding
- Kayaking
- 4x4 vehicle tours

Educational

- Environmental education
- School field trips & science projects
- Team-building ropes course

Soft Adventure

- Self-guided exploring (Letterboxing or Orienteering)
- Primitive skills survival training
- Ranch tourism
- Conservation Ranch Photography (Appendix B)*
- Luxury tent ecolodge (Appendix D)¹

¹ Field work has not yet been completed to provide details of these activities and if in fact they are viable opportunities for Hacienda Santa Monica.



Example of 4x4 that could be used for ecotourism at HSM.



Whimbrel walking along the coastal edge.

VIII. RECOMMENDATIONS

The following recommendations can be applied to develop ecotourism in this site:

1. Conduct more wildlife and bird surveys during all seasons based on a scientific methodology: More in-depth studies/surveys need to be conducted to determine species that can be observed here, frequencies, abundance, time of year, etc. A bird and wildlife checklist, and eventually field guides for purchase, should be developed. We observed more than 100 bird species during this short assessment which indicates that bird watching could play a major role in ecotourism development at HSM. See Appendix B for more details on bird observations at HSM.
2. More fieldwork is needed to assess the potential for nature-based tourism in the Juan Hombrón Unit, especially the estuary and extensive mangrove forests of the western end: This area has the highest potential for nature-based tourism, but it must be explored more extensively. Wildlife surveys must be conducted, possible kayak routes scouted and possible sites for luxury tent ecolodges located.
3. Consider conservation of biodiversity in all aspects of the master plan: Develop a plan for green infrastructure, preservation of green spaces and wildlife habitat and conservation of forests. The large Hacienda Unit is mostly devoid of trees, a reforestation plan using native species, should be developed for this area. See Appendix B.
4. During follow-up field surveys, explore opportunities for birding tourism, luxury tent lodging, conservation photo tours, and guided tours with four-wheelers: All of these activities have the potential to attract high-end tourists, both local and international. For preliminary information on these activities see Appendices B, C, and D.
5. Investigate and understand the importance of the remnant tropical (also known as Panamanian) savanna ecosystem: A large part of the remnant Panama Savana is found in the HSM area. This ecosystem is very important for wildlife and plant biodiversity conservation and sustainability. In-depth studies should be conducted in this area and a conservation plan specific to this ecosystem should be developed.
6. Follow principles of benefits-based tourism and international standards for tourism development in wetlands & mangroves.

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Appendix A

Sustainable Tourism and Biodiversity Conservation Model

This is a model of sustainable tourism development that provides linkages and synergies between visitors, communities, and the environment, such that benefits in one area create benefits in the other two.

Conservation relates to the overall health of the environment, as measured by biodiversity and preservation of historic sites. It also ensures high-quality settings for residents and visitors and the product base for tour providers. Indicators include:

- Acres/hectares protected or restored
- Historic and cultural sites protected or restored
- Scenic vistas enhanced
- Number and types of species observed
- Reduction of depreciative behavior (litter, looting, wildfires, graffiti, etc.)
- Political support for conservation agencies and programs
- Hours and money donated
- Entrance fees collected

Community and Economic Development enhance the quality of life for residents and create business opportunities. Successful tourism businesses and communities provide infrastructure and services, financial and volunteer support for conservation projects, and political support for conservation projects and agencies. Benefits include increases in:

- Numbers of jobs related to conservation
- Economic diversification and percent of jobs conservation-related
- Income
- Profits
- Taxes
- Infrastructure for locals and visitors

Quality Visitor Experiences are the foundation for successful tourism. They depend upon properly managed, resources, settings and attractions (including scenery), professional tourism services and infrastructure, and adequate visitor information and interpretation. Without a quality visitor experience there will be no sustainable tourism and no public or financial support for parks and conservation. Potential benefits include:

- Increased customer satisfaction and loyalty
- Longer stays and willingness to pay
- Repeat visits
- Word-of-mouth advertising
- Positive psychological, social, and physiological changes for the individual
- Education and appreciation
- Changes in visitor attitudes and behavior



Sunrise over Hacienda Santa Mónica.



Benny Wilson and Wayne Arendt surveying birds in the Juan Hombrón Unit.

Appendix B

Areas within Hacienda Santa Mónica conducive to Eco- and Avetourism, and ways to enhance their attractiveness

Two avian experts, a USFS-IITF Research Ornithologist (Wayne J. Arendt) and one of Panama's foremost eco- and avetourism guides (Benny Wilson), surveyed Hacienda Santa Monica over a two-day period. More than 100 species of birds from diverse taxonomic groups (hummingbirds to birds of prey) were observed.

Following is a list of strong candidate sites that will appeal to ecotourists, international birders, and nature enthusiasts in general; also included are recommendations for increasing local and regional plant and animal diversity, while improving the Hacienda's green infrastructure:

Tropical Savanna

- A number of high profile, flagship bird species were observed throughout the sprawling savanna habitat. Two species are especially illustrative of the national and international importance of this fragile ecosystem. A Long-winged Harrier (*Circus buffoni*) was observed flying over the grasslands on 20 March 2015. It is only the second record for Panama. Although the species is currently listed as a vagrant (2 records), it is likely that its numbers will continue to increase, along with those of many other species, as a result of global climate change. The harrier's historical range included several countries in South America (Argentina; Bolivia, Plurinational States of; Brazil; Chile; Colombia; French Guiana; Guyana; Paraguay; Suriname; Trinidad and Tobago; Uruguay; Venezuela, Bolivarian Republic of; see: <http://www.iucnredlist.org/details/22695373/0>).
- Contingent upon recent DNA testing (results still pending), Panama's population of the Yellow Grassland Finch (*Sicalis luteola*) may become a separate species. If this occurs, the species will immediately be eligible for classification as *Critically Endangered* because it will be at high risk of global extinction, owing to its small population and remnant habitat in Panama. Although an all-inclusive species and habitat preservation and management program would have to be put implemented beforehand, this area would become a "mecca" for international birders, especially 'listers' (known as 'tickers' in the UK) and everyone else, from near and far, who have a keen interest in nature and observing newly recognized species.
- Characterized by its multifunctionality, the green infrastructure within the HSM property and its surroundings consists of a network of natural green spaces, e.g., greenbelts (forested or other vegetation association buffers surrounding agricultural areas), greenways (ecological grassland corridors), hedge rows (low-to-medium height bushes) and 'live' fences made of growing trees and cacti, acting synergistically to increase vegetative connectivity, which directly translates into increased floral and faunal biodiversity.
- One of the weaknesses of the HSM property overall is its lack of trees, the planting of which as greenbelts around agricultural plots or as shade overstory in the agricultural areas would greatly help to lower

the ambient temperatures and increase humidity in the resultant microhabitats they would form throughout the hacienda during the dry season and extended periods of drought, which are forecasted to continually increase throughout the region in the near future.

Rivers, streams, freshwater swamps, saltwater marshes, drainage ditches and water-catchment basins (small ponds) scattered throughout the hacienda, including the areas under cultivation.

- Several species, and at times hundreds of individuals, of migratory and resident waterfowl, shorebirds, and landbirds were observed in these areas. Many of the observed species are in decline and/or are species of global conservation concern. Thus, as with all of the wetland areas surrounding the Panama Bay, the wetlands, including agricultural areas, contained within HSM are of great importance as: (a) stopover destinations for passage migrants; (b) “wintering” grounds for seasonal migrants, which often remain as long as nine months of the year; (c) breeding sites for a diverse group of resident birds; and (d) refugia for all avifauna and other plant and animal species.
- To enhance the quantities and diversity of birds in these areas, contamination of the various wetland types from pollution by biocide runoff from agriculture, etc., must be reduced; contaminant studies are recommended.
- Restoration of native wetlands vegetation in all of these areas
- Planting of greenbelts

Coast and Mangrove forest

- Ideally, ecotourists visit sites to observe wildlife and as a result spend money in the area. Well-planned ecotourism can provide economic and political incentives for conservation of these two areas, and could bring additional benefit to the local community and the regional economy.
- Birding and mangrove exploration is a popular form of ecotourism, an activity that may ultimately help HSM improve its management of the natural resources on the properties.
- The large wetlands area adjacent to the coast harbored dozens, and at times hundreds, of migratory ducks, terns, and a wide array of wading and marsh birds, many of which are of global conservation concern.

- Many of the attributes and enhancement recommendations included above are germane to these areas.

- Owing to its proximity to Panama City, HSM, including the coast and mangrove forest, can be reached with minimal travel efforts.
- As documented elsewhere in this report, to attract more visitors to these important natural areas, it is imperative that additional accessibility infrastructure (boardwalks, viewing towers) be constructed and well maintained.
- To bolster the local economy, we recommend the use of local guides because nature based tour operators must possess a strong knowledge of, and an even stronger affinity for, these natural areas and incumbent plants and animals. For this reason, selection of the appropriate, local ecotourism guide is essential.
- Owing to the expected high volume of visitors, we recommend a buffer zone around these two areas to protect not only wildlife but also the unique vegetation.

Social Impacts

- In addition to economic and environmental impacts, ecotourism can have social effects.
- Ecotourists could use small boats or kayaks to view the wildlife in the wetlands area. The local economy would also be augmented if the boat operators are from the surrounding communities.

Harassment of wildlife

- Ecotourism, in all of the areas under discussion, has the potential to cause harmful environmental impacts. One issue is the harassment of wildlife. For example, feeding the birds and other wildlife can create unnatural behavior, which can be dangerous to their wellbeing. As a case in point, some birders attract birds by producing sounds (e.g., “pishing,” hand squeaking, and whistling) or by playing a tape-recorded song, which brings the birds out into the open, thus causing undue stress and exposure to predators.
- Ecotourists can set fire to the mangrove forest, which could have devastating impacts on the vegetation and wildlife habitat.
- Another issue is the trampling of the vegetation. Ecotourists often venture off the trail in pursuit of birds and other animals, damaging the underlying vegetation and soil in the process. Land stewards will need to use discretion in addressing such issues.

Appendix C

Conservation Ranch Photography, A New Model for Sustainable Development in Panama

In the last few years there has been an effort to combine conservation of natural resources with nature photography, to what is now termed *Conservation Photography*. This new concept has largely been fueled largely by a creative idea to use photography as a tool to help preserve wildlife habitat and to generate income for private landowners, mostly in the US southern Texas *hill country* where economic challenges were forcing landowners to sell ranches that had been in their family for generations. These large tracts of land, whose economic base was cattle ranching, also provided habitat for wildlife. Although, at the time, wildlife habitat conservation was not a major focus for the landowners and it did not provide economic benefits to them.

Also, about this time, the photography industry was in a mode of transformation from film to digital, making photography more attractive and accessible to millions of

persons who otherwise would probably not be interested in photography. As more cameras were sold to amateur and hobbyist photographers, opportunities arose to provide training programs to this growing population of interested persons, who also had time and financial resources to pay for high-end camera equipment and training. Professional photographers and photography trade associations in the US keyed in on this new opportunity and began promotion and marketing of high quality photography books and other materials. This generated more interest and more opportunities and the demand for more photography field training programs.

Thus, a novel idea was put forward by John Martin, a ranch owner in Texas, i.e. to combine conservation and nature photography as *Conservation Ranch Photography*. In 2003 Mr. Martin founded an NGO “Images for Conservation



◀ Howler monkey resting in riverine forest.yhh

▲▲ Example of conservation photographers (awindowintothewoods.com).

▲ Camouflaged wildlife photographer (www.beebower.com)



Permanent photography blind and artificial waterhole (source: www.stephenfisherphotography.com), inset conservation photographers (source: www.santaclararanch.com).

Fund (ICF) [<http://imagesforconservation.org/>] for the purpose of “conserving wildlife habitat and generating a sustainable income stream within rural economies through the establishment and prosperity of the Private-Lands’s Nature-Photography Industry (PLNPI)” (ICF web page). Thus a new industry was formed; and today the ICF provides a valuable service for landowners, nature photographers and, most importantly, wildlife.

The ICF sponsors an annual *Pro-Tour of Nature Photography* contest, where it pairs a landowner with a professional photographer for one month. Several teams are organized each year on several ranches. Working together, the photographer-landowner produce a large volume of high quality photographs for the contest. The winning photographs are then used to produce a coffee table book and for conservation efforts by the landowners, the photographers, the ICF, local government organizations, and for environmental education purposes. In addition to this professional photography contest, the landowners also develop *photo opportunities* on their land, allowing any photographer to use their land as a photographic field station. Of course the photographer pays a fee for this, and in most cases, it is required for the photographer to hire a local guide. Most landowners also have small, bed-breakfast lodging available for the photographer, producing an excellent service to the photographer and economic benefits to the ranch and the local communities.

The ICF also has other programs for amateur photographers, for youth, and for schools. Are are focused on wildlife

habitat conservation and economic benefits for local communities.

The ICF example can be used by HSM to develop a similar model for Panama. Many of the attributes that have proven successful in the Texas hill country *Conservation Ranch Photography*, are also present at HSM (such as a large tract of single-ownership land, mega/unique wildlife, flagship bird species, luxury accommodations nearby, visitor safety, etc).

Formation of a conservation photography program developed by HSM, just as with ICF, can help to:

- provide a solution to assist with biodiversity conservation.
- promote responsible land stewardship.
- provide the unique ability to contribute to biodiversity conservation and responsible land stewardship by making it profitable to conserve habitats permanently.
- provide sustainable use of private land as a ‘win-win’ solution for wildlife and for regional owners who face increasing pressure to develop their lands for commercial uses.
- involve local communities in conservation efforts and provide for local economic benefits.
- become a model for other private landowners to follow in the HSM area, as well as, other locations in Panama.
- establish a new tourism niche industry in Panama.

Appendix D

Luxury Tent Ecolodges

A little used option for sustainable green development in Central America is “luxury camping” or “deck camping” tent ecolodges. These are canvas and/or fabric structures, usually placed on a wooden platform on land or over water, that have a small footprint and low environmental impact. Widely used in Africa, this concept of sustainable lodging is just catching on in the Americas. The Juan Hombrón Unit offers ideal conditions for such sustainable development, where comfortable lodging can be provided to clients wishing for an outdoor or camping experience, but at the same time, enjoy luxury, glamor, and adventure in a safe and beautiful landscape.

At the Juan Hombrón Unit, this luxury lodging experience can be combined with high-level bird watching, kayaking in the estuary, beach combing, hiking and just general enjoyment of solitude and the natural environment, as well as, *Conservation Ranch Photography* (see Appendix C). This alternative offer is a quite different experience from that offered in nearby cement resorts and the *sand & sea* tourism.

For more specific information, designs, examples from other areas, etc. conduct an internet search for luxury lodging, tent camping, deck camping, sustainable camping, tent-cottages, and similar keywords and a wealth of information will be available. Some specific sites include www.coloradoyurt.com, www.exclusivetents.com, and similar sites.



Examples of luxury tent ecolodges. Photo sources:

<http://www.asctents.com>

<https://secure.ifai.com>

<http://www.indochine-safari.com>

<http://ctents-luxury.blogspot.com>

<http://www.worldpropertyjournal.com>

